School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Scribe 3

Group 1

Section 2

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Scribe Question 1: System Objective and Deterministic Baseline

Clearly restate your project objective. Identify and describe the deterministic baseline strategy you are implementing. Explain why this baseline is reasonable or commonly used in your domain.

Our project aims to create a simulation of a four-way intersection to study how probabilistic modelling helps in reducing congestion brought about by stochastic (random) traffic flow. The primary goal is to minimise the waiting time/delay of cars at the intersection, i.e., the time cars spend at a red light.

The deterministic baseline model that we implemented is the "dumb light": the traffic light follows a predefined cycle (e.g., cycle length C is always the same) where each direction is given a predetermined amount of time (e.g., North-South is given g_{NS} seconds of green time, then East-West is given g_{EW} seconds of green time, and this cycle repeats indefinitely regardless of the length of the queues or the randomness of the incoming traffic). The decision is completely deterministic; given the time of day within the cycle, the next phase is completely predetermined.

The fixed time controller is a basic model in traffic systems because it is simple, predictable, easy to implement, and commonly used in real-world intersections when sensing or actuation is limited or when engineers use historical averages for timing plans. It is also a useful baseline to test whether probabilistic modelling and adaptive logic reduce waiting time under uncertainty.

Scribe Question 2: Random Variables in the Simulation

Identify the key random variables in your system. For each variable:

- **Provide a formal definition.**
- **Specify the assumed distribution.**
- **Explain how it is used within the deterministic baseline simulation.**

Note: The decision rule must be deterministic, even if uncertainty exists in the system inputs.

The key random variables in our system are as follows:

1. **Vehicle Arrival:** For lane/approach i, let

   $$N_i(\Delta t) = \text{number of vehicle arrivals in lane } i \text{ during the time interval } [t,\, t + \Delta t]$$

   Distribution Assumption:

   $$N_i(\Delta t) \sim \text{Poisson}(\lambda_i \Delta t)$$

   where λ_i is the average arrival rate for approach i.

   At each simulation time step, we sample $N_i(\Delta t)$ and add that many cars to the queue of lane i. However, the signal timing does not react to this; it continues its fixed cycle.

2. **Interarrival time between two consecutive vehicles:**

   Let T_i be the time gap between 2 consecutive arrivals on approach i.

   Under a Poisson process:

   $$T_i \sim \text{Exponential}(\lambda_i)$$

   This is used when we simulate continuous-time arrival spacing to produce realistic "gaps" and arrival clustering behaviour under the Poisson assumption (the fixed time controller remains unchanged)

3. **Vehicle Speed**

   For each vehicle j, let V_j be the speed of that vehicle.

   $$V_j \sim \mathcal{N}(\mu, \sigma^2),$$

   To represent the real world variation between drivers and vehicles.

When an approach is green, sampled speeds determine how many cars can pass during that green, This affects how fast queues shrink, even though the green timing is fixed.

4. **Queue Length**

   Let $Q_i(t)$ be the no. of vehicles waiting in lane i at time t.

   $Q_i(t)$ is assumed to be a random process with random arrivals and random service.

   It is updated step by step:

   1. Firstly when arrivals increase the queue
   2. Secondly when during green light, when the queue decreases.

   The baseline does not directly use $Q_i(t)$ to decide green; however, we still record as a major outcome

5. **Waiting time per vehicle**

   For a vehicle j, let W_j be the waiting time spent by a vehicle before the intersection is cleared.

   W_j is an output random variable determined by the stochastic queue evolution

   We compute W_j for each vehicle and summarize it via averages and distribution plots (PDF/CDF) to evaluate baseline performance.

Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule

Explain how uncertainty is modeled in your simulation while keeping the decision strategy deterministic.

- **What probabilistic assumptions are being used?**
- **How does the deterministic baseline operate under uncertain inputs?**
- **Provide relevant mathematical expressions if applicable.**

The source of uncertainty are the system input and dynamics (in terms of arrivals and speeds), but the decision rule (the signal phase selection and the green time allocation) is fixed and does not involve any random elements or observations of the queue. In other words, the randomness is in the evolution of the intersection, not the decision process.

(A) Poisson Arrival for each approach:

$$N_i(\Delta t) \sim \text{Poisson}(\lambda_i \Delta t), \qquad P\big(N_i(\Delta t) = k\big) = \frac{(\lambda_i \Delta t)^k e^{-\lambda_i \Delta t}}{k!}$$

(B) Exponential inter-arrival gaps:

$$T_i \sim \text{Exponential}(\lambda_i)$$

(C) Normal speed variability:

$$V_j \sim \mathcal{N}(\mu, \sigma^2)$$

How the baseline reacts to uncertain inputs:

Let the signal have a fixed cycle C and fixed green allocations (g_{NS}, g_{EW}). Define a deterministic phase function:

$$\text{phase}(t) = \begin{cases} NS & \text{if } t \bmod C \in [0, g_{NS}) \\ EW & \text{if } t \bmod C \in [g_{NS}, C) \end{cases}$$

This is deterministic because it depends only on time, not on random arrivals, speeds, or queues.

Randomness still affects outcomes because the queue evolves as:

$$Q_i(t + \Delta t) = \max\big(0,\ Q_i(t) + N_i(\Delta t) - D_i(\Delta t)\big),$$

where $D_i(\Delta t)$ (departures) depends on whether that lane is green and on the sampled speeds (service capability). The controller never changes g_{NS}, g_{EW}; it just runs the schedule and we measure queue/waiting-time performance under randomness.

Scribe Question 4: Simulation Pipeline

Describe the complete simulation pipeline.

- **Input generation**
- **Processing stages**
- **Deterministic decision mechanism**
- **Output metrics**

Clearly explain how the full system operates.

The simulation pipeline explains the step-wise operation of the entire traffic system that begins with the creation of traffic to the creation of the final performance metrics

i) Input Generation

The Generation of the input traffic is the first step in the simulation. Real traffic is random and, therefore, we would simulate this randomness through probabilistic models.

The arrival of vehicles at each lane of the four way intersection is produced in the form of a Poisson process. This will imply that the arriving number of cars in a short period will follow:

$$N_i(\Delta t) \sim \text{Poisson}(\lambda_i \Delta t)$$

where λ_i represents the average arrival rate for lane i.

To ensure the traffic flow is more realistic, the time separation between the successive vehicles is also modelled by exponential distribution:

$$T_i \sim \text{Exponential}(\lambda_i)$$

As well, the vehicles are given a speed randomly chosen out of a normal distribution:

$$V_j \sim N(\mu, \sigma^2)$$

This is the fact that various drivers have different speeds.

ii) Processing Stage

After the vehicles are created the simulation then processes them one step at a time. Each time the simulation process runs:

- New vehicles are introduced to the queue of each lane according to the arrivals sampled.
- There is a queue length of each lane $Q_i(t)$ meaning the number of cars that are waiting at this time.
- With the red signal, the queue increases as cars continue to come in, yet they are unable to move out.
- When the lights are green then the cars start clearing the crossing after their speeds.

The stage is simulating the way in which traffic queues build up and depreciate.

iii) Deterministic Decision Mechanism

Although the arrivals at the traffic are random, the signal control is deterministic. The traffic light is operating on a regular schedule:

- The green for North-South direction is given for g_{NS} seconds.
- The green for East-West direction is given for g_{EW} seconds.

The time is the only basis of the signal phase:

iv) Output Metrics

The primary product of the simulation is the anticipated waiting time of vehicles.

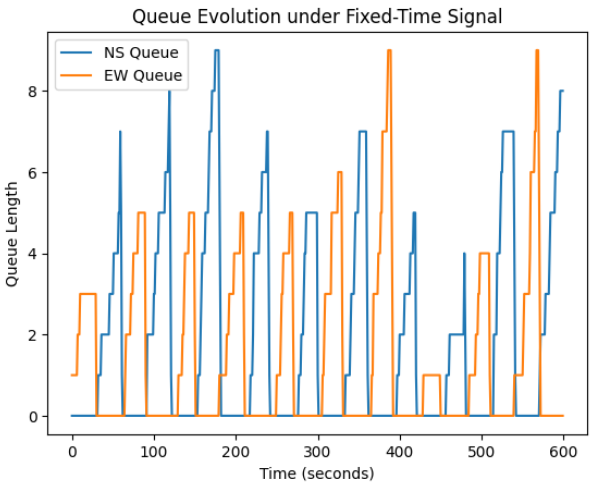
The delay at the queue is calculated individually to every vehicle. Such delays are added to give the system performance.

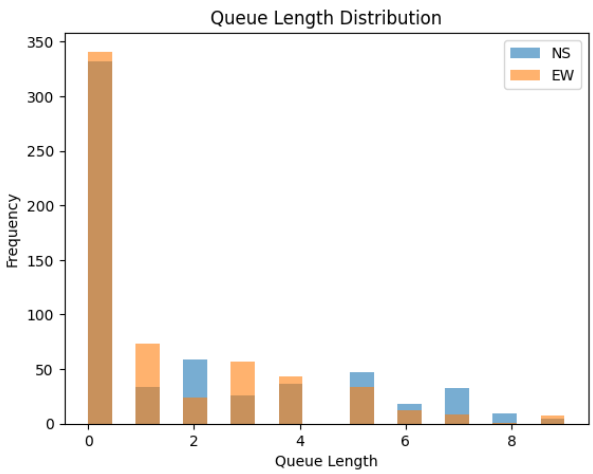
The key performance objective is: Reduce Expected Waiting Time $E[s]$.

Other statistics are also registered like the length of the queue and delay per vehicle.

Lastly, these results are applied in policy evaluation where deterministic baseline analysis is evaluated and compared to future strategies.

The traffic system is modelled with the help of a discrete-time simulation. The arrival of vehicles in each lane is determined by a Poisson distribution and vehicles become part of queues at the intersection. The traffic light is of facilitated time-lead cycle wherein every direction is given a predetermined period of green light. Vehicles are sampled through normal distribution to give the number of cars that pass through the time of green, and the lengths of queue are updated with the time.





Scribe Question 5: Simulation Outputs and Interpretation

Present and interpret the results obtained from your deterministic baseline simulation.

- **What performance metrics are used?**
- **Have PDF/CDF plots or other statistical summaries been generated?**
- **What patterns or behaviors are observed under the deterministic strategy?**

We interpret the output of the system after completing the deterministic baseline simulation to get an idea of how the system behaves in terms of traffic of fixed signal timing.

Performance Metrics

The main measure employed is expected waiting time also known as the average time spent by vehicles in the intersection.

For each vehicle we measure:

- The delay at the intersection point.
- Queue length experienced
- Delay accumulated in total in the system.

It is these values that enable us to assess the efficiency of the traffic signal policy.

Statistical Summaries (CDF and PDF)

To study the behavior of the system better, a number of statistical distributions are produced.

The simulation results are Probability Density Function (PDF) and Cumulative Distribution Function (CDF) plots.

Examples include:

Arrival Distribution (PDF)

Demonstrates the likelihood of varying numbers of cars within a time frame. The distribution takes the shape of the most frequent arrival counts, as it is expected in a Poisson process.

Arrival CDF

Demonstrates the likelihood of the number of cars that arrive being less than some specific number.

Speed PDF

The normal distribution would imply that the driver speeds would be in a bell-shaped curve with the mean of 40 km/h being the average speed.

Speed CDF

Displays the cumulative probability of vehicles with speeds that are below a given threshold.

The plots also assist in validating that the simulation inputs are as intended (probabilistic models).

Observed Behaviour

The findings indicate that there exist a number of patterns under the deterministic signal policy.

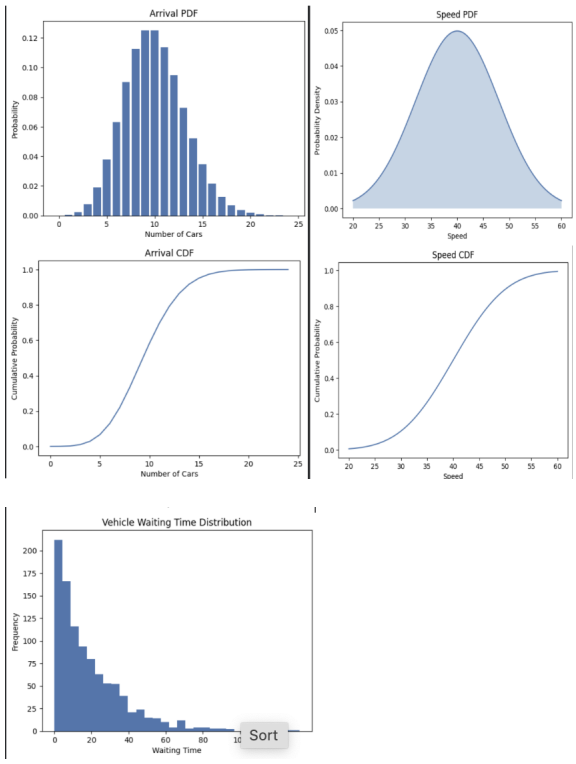
First, there is the accumulation of queues in case of a large number of vehicles arrival during a red signal.

Second, the timing of the signal is preprogrammed, hence the system is unable to deal with sudden traffic surges. This may cause an increase in waiting period in the lanes.

Third, it has been observed that in some cases; a lane is allocated green time although it has few vehicles using that lane and this wastes capacity.

The behaviours point to the weakness of fixed-time control in stochastic traffic environment.

The generated data is analysed with the help of statistical graphs after running the simulation. To monitor the distributions of vehicles arrivals and speed, PDF and CDF graphs are generated. The patterns of congestion as time is also plotted under the deterministic signal policy to observe the queue length over time.



Scribe Question 6: Technical Gaps and Preparation for Next Stage

Identify current limitations of the deterministic baseline.

- **What aspects of the model are simplified?**
- **Where might randomness improve performance in future milestones?**
- **What refinements are planned for Milestone 4?**

Despite the fact that the deterministic baseline is simple to start off with, it is limited in a number of ways.

Current Limitations

The existing model has a number of weaknesses.

Non-reactive signal policy

The signal is not changing the signal according to the length of the queue and the state of the traffic.

Static timing inefficiency

The 30-second fixed phases can be inefficient in cases where the traffic demand varies.

No spillback control

The model is not yet designed to accommodate the experience of long queues that are blocking upstream lanes.

Due to these constraints, the deterministic basis can be a bad actor in periods of a traffic spike.

Improvement Opportunities.

The milestones in the future will strive to come up with smarter decision strategies.

The system would also be able to dynamically change the phases of the signals according to the circumstances of real-time traffic, instead of being operated according to a fixed schedule.

The potential enhancements are:

- Adaptive green times depending on the length of queues.
- Dynamic phase selection
- Probabilistic or stochastic decision making.

These strategies will help minimize the average waiting time.

Refinements that will be made to Milestone 4.

In Milestone 4, the project shall be gravitated towards more intelligent signal control approaches.

- The planned improvements are:

- Moving to smarter decisions logic.
- Dynamic phase selection in terms of weight.
- Markov state transition signal control models.

These improvements should help the traffic light to be more sensitive to the stochastic environment and enhance the performance of the intersection.